

SANCHARIKA DEBNATH

Applied AI Engineer

Applied AI Engineer with 3+ years building production LLM systems, RAG pipelines, and AI automation across defense, logistics, and SaaS. Experienced deploying offline GenAI in air-gapped environments and migrating cloud infrastructure at scale.

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WORK EXPERIENCE

Graduate Apprentice (CSE)

Defense Research and Development Organisation (DRDO), Odisha, India

Jan 2025 – Aug 2025

- **Deployed an air-gapped LLaMA/Mistral inference stack** across Linux and Windows in a high-security network with no internet access, enabling 12+ researchers to run AI queries locally — reducing average research query turnaround from approx. 2 hrs to under 35 min (approx. 70% reduction).
- **Eliminated manual tender drafting** for a procurement team processing 20–30 documents/week by building a Tesseract + EasyOCR extraction pipeline feeding a LangChain/Ollama RAG system that outputs structured, compliance-ready LaTeX tender documents directly from noisy scanned PDFs.
- **Mentored a cohort of 10 interns** on their internship project, providing technical guidance, unblocking implementation issues, and ensuring delivery within the program timeline.

AI Product Developer (concurrent freelance)

Leapon, Remote

Feb 2024 – Present

- **Built a prompt-orchestrated voice AI agent** using ElevenLabs and a custom backend, enabling 3+ enterprise clients to automate inbound/outbound call flows with configurable tone, persona, and conversation logic — replacing manual handling for repeatable communication workflows.
- **Migrated full production stack** from AWS (RDS MySQL + S3) to GCP with under 60 min of downtime, cutting monthly infrastructure costs by 46% (\$130 → \$70/month).
- **Automated business card digitization** using Google Vision API OCR, parsing unstructured card images into structured vCard and Excel output — eliminating manual data entry for a sales team's CRM onboarding process.

AI/ML Engineer

Giggr Technology, Karnataka, India

Jun 2023 – Feb 2024

- Built an event management system integrating facial recognition for attendance, object detection via camera, and GPS tracking. Achieved **87.4% accuracy** using **OpenCV, YOLO, DeepFace, and Geolocation API**.
- Delivered a public API via **Elasticsearch** covering 1.3M+ Indian education institutes using web scraping and **AWS Shell Scripting**; co-developed **Neo4j** graph database architecture for data mining and analytics.
- Built a **Dialogflow + GPT-powered** chatbot with Firebase integration and advanced scoring logic, increasing user engagement sessions from approx. 800/month to 1,500/month within the first quarter post-deployment.

Backend Developer

Leapon, Remote

Feb 2023 – Jun 2023

- Built a customized Django REST API backend for a platform with 50+ advisors, with unit testing, bug fixing, and CI/CD integration. Stack: **Python/Django, SQL, AWS, Git, JIRA**.

Data Science Intern

Maersk Global Service Centres, Karnataka, India

Jul 2022 – Feb 2023

- Clustered global port data using **Gaussian Mixture Models** for cost estimation and modeled container turn times using **XGBoost and Regression** to forecast attachment ratios, contributing to a **15% increase in freight profit**. Stack: **PySpark, Databricks, Microsoft Azure**.
- Applied **K-Means clustering** to segment large-scale customer datasets, identifying behavioral patterns to optimize digital acquisition and personalization strategies.

TECHNICAL SKILLS

Core: Python, LangChain, Ollama, FastAPI, RAG Architectures, LLMs, GenAI

Proficient: PyTorch, TensorFlow, OpenCV, FAISS, Neo4j, GCP, AWS, MySQL, Tesseract, HuggingFace

Familiar: PySpark, Azure Databricks, Elasticsearch, scikit-learn, Java, R, JavaScript

Practices: NLP, Computer Vision, Deep Learning, REST API design, CI/CD, Agile/Scrum

PROJECTS

[GenieFile](#)

Graph-Augmented RAG Document Assistant | Python, LangChain, FAISS, Neo4j, Ollama

- Built an end-to-end RAG pipeline supporting PDF, DOCX, and TXT ingestion at scale, combining **FAISS dense vector search** with a **Neo4j Knowledge Graph** for cross-document entity relationship extraction.
- Implemented a custom eval framework tracking Retrieval metrics (Precision@K, Recall), Generation quality (groundedness, hallucination rate), and system latency to iteratively optimize LLM responses.

[LLaMA 3 from Scratch](#)

Transformer Architecture Implementation | Python, PyTorch, Meta weights

- Implemented the full LLaMA 3 architecture from scratch using PyTorch — covering tokenization, positional embeddings, multi-head attention, RoPE, RMSNorm, and feedforward layers — loaded against official Meta model weights to verify correctness.
- Built to develop ground-level understanding of transformer internals; capable of end-to-end text generation without any HuggingFace abstractions.

[Signature Verification System](#)

Siamese Neural Network | Python, PyTorch, ResNet, Spatial Attention

- Designed a **Siamese Neural Network** with a **ResNet backbone and spatial attention mechanism** to distinguish genuine vs. forged handwritten signatures by comparing deep feature representations.
- Built end-to-end pipeline covering data preparation, contrastive loss training, and threshold-based verification evaluation applicable to fraud detection and identity verification domains.

[Career Enchanter](#)

GenAI Resume Intelligence Pipeline | Python, BERT, Gemini-Pro, Django

- Developed an automated career intelligence platform using **BERT Transformers** and **Gemini-Pro** to ingest unstructured resumes, compute ATS compatibility scores, and generate targeted improvement recommendations.

PUBLICATIONS

- [Hyperspectral Image \(HSI\) Compression and Classification](#) — Research framework for HSI classification using a CNN bottleneck AutoEncoder with Hyper Spectral Net. Achieved **0.998 accuracy**. Stack: **TensorFlow, Autoencoder, Hybrid SN**. Published in **IJCISIM** (Mar 2022).

EDUCATION

Kalinga Institute of Industrial Technology

B.Tech, Information Technology | CGPA: **8.83/10** (Distinction)

Bhubaneswar, Odisha, India

Jun 2019 – Jul 2023